CS/CNS/EE 156b - Learning Systems

Tips for CheXpert Challenge

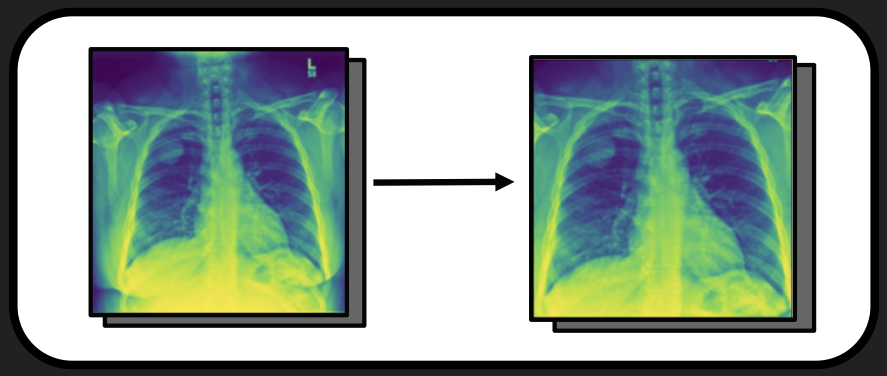
Data Preprocessing

Image Format

- How large are the radiograph images provided in the dataset?
- Should the images be resized?
- Are the images colored? Is coloration important information in this context?

Image Augmentations

- Some other properties of the radiographs you should consider include:
  - Zoom
  - Brightness
  - Image orientation (translation, rotation)
- Normalize? Artificially introduce variations?
- At the same time, consider the tradeoff that comes with training on augmented data:
  - Training time
  - Robustness



Dataset Balance

- Consider the balance between negative, positive, and uncertain examples for the nine pathologies in your training dataset.
- How can we offset imbalance in the training set?
  - Manual rebalancing by selecting X amount of images from each class
  - Using a GAN to generate more examples for data-limited classes
- The second option is interesting, but also consider whether or not there is enough data to train a GAN and create realistic examples.

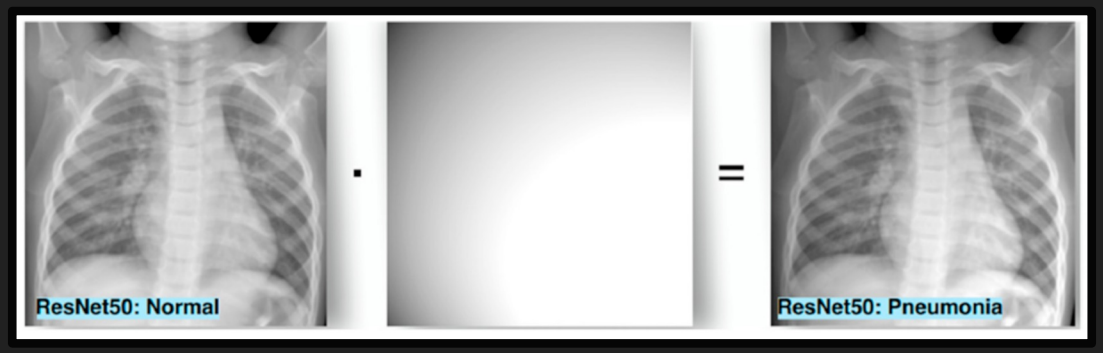
Pathology	Positive (%)	Uncertain (%)	Negative (%)
No Finding	16627 (8.86)	0 (0.0)	171014 (91.14)
Enlarged Cardiom.	9020 (4.81)	10148 (5.41)	168473 (89.78)
Cardiomegaly	23002 (12.26)	6597 (3.52)	158042 (84.23)

Considering “NaN” Data

- Some radiographs are not labeled, and instead have “NaN” values for one or more pathologies. How will you account for this?
- Simple options:
  - Disregard the “NaN” data
  - Replace “NaN” data with negative or uncertain labels
- Semi-supervised learning to predict labels from the labeled radiographs?

Adversarial Robustness

- Consider some variations in the radiographs that may not necessarily be covered by adding image augmentations in training dataset.
- These variations can come in the form of visual cues such as blur, but are sometimes not apparent to the naked eye.
- Variations can contribute to your model learning to identify features based on the *wrong signal*.
- Can teach the model to identify invariant features through the use of adversarial examples.



Feature Extraction

What to look for

The raw image data can be processed with a NN, but this will likely only go so far.

To help out your model, you can give it more information to work with.

What may change in an image if there is a fracture? What about pneumonia?

To help answer these questions you can extract features from the data and work with those

Possibilities

Some things to look for

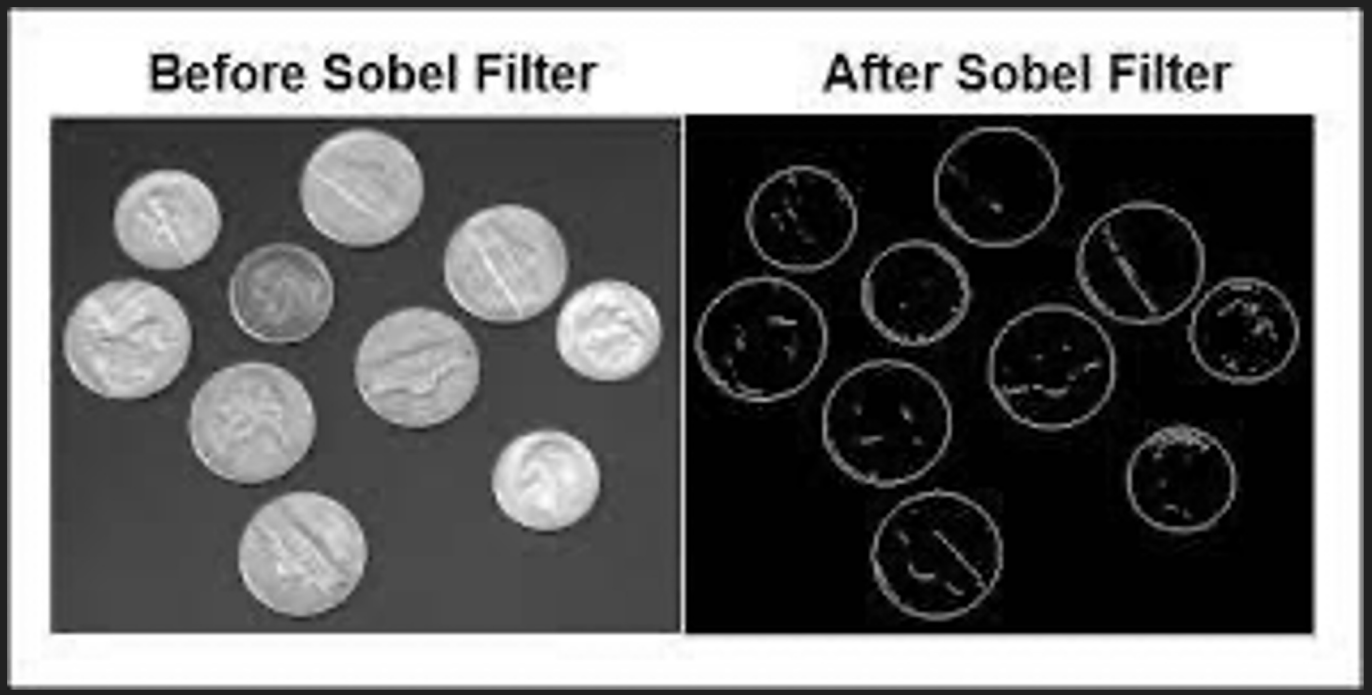
- Pixel intensity
- Symmetry
- Edge detection

Methods you can use (very non exhaustive)

- RGB
- Filters (Sobel, Prewitt, etc.)
- Kernels
- Use Histograms! Use scatter plots!

Example of a Sobel filter

Edge detection:



Building off of intuition

There is a large solution space for things you can extract, you should guide what you pull from images based on intuition

From the classes that we are working with, what should appear in images that have these pathologies?

Some may be more intuitive than others, to get started let's consider an easy one

Fracture

Lung opacity

Pneumonia

Pleural Other

Enlarged Cardiom.

Cardiomegaly

Support Devices

Pleural effusion

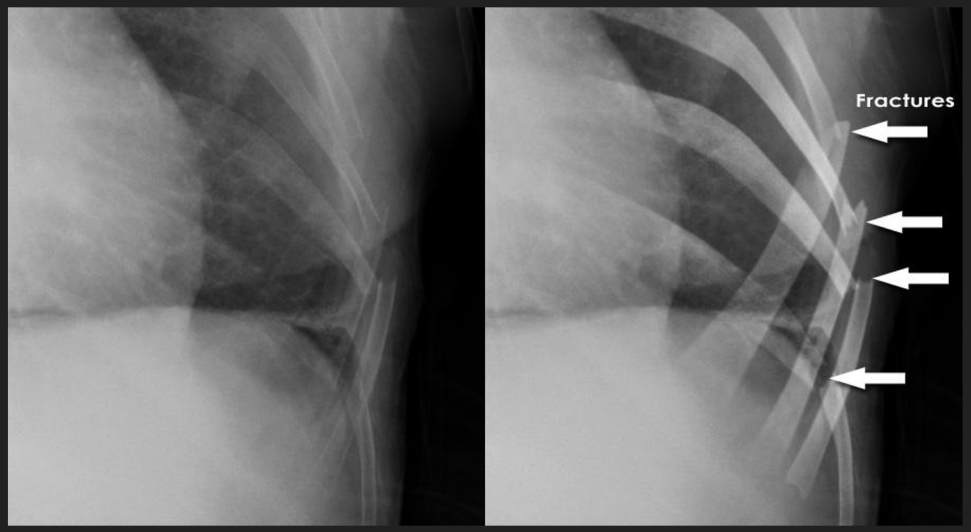
Fractures

This will be one of the features that may be best suited for feature extraction

Fractures will often (not always) be visible and result in asymmetry and discontinuity of shapes and edges.

So extracting symmetry, shapes,

and edges will help you classify.



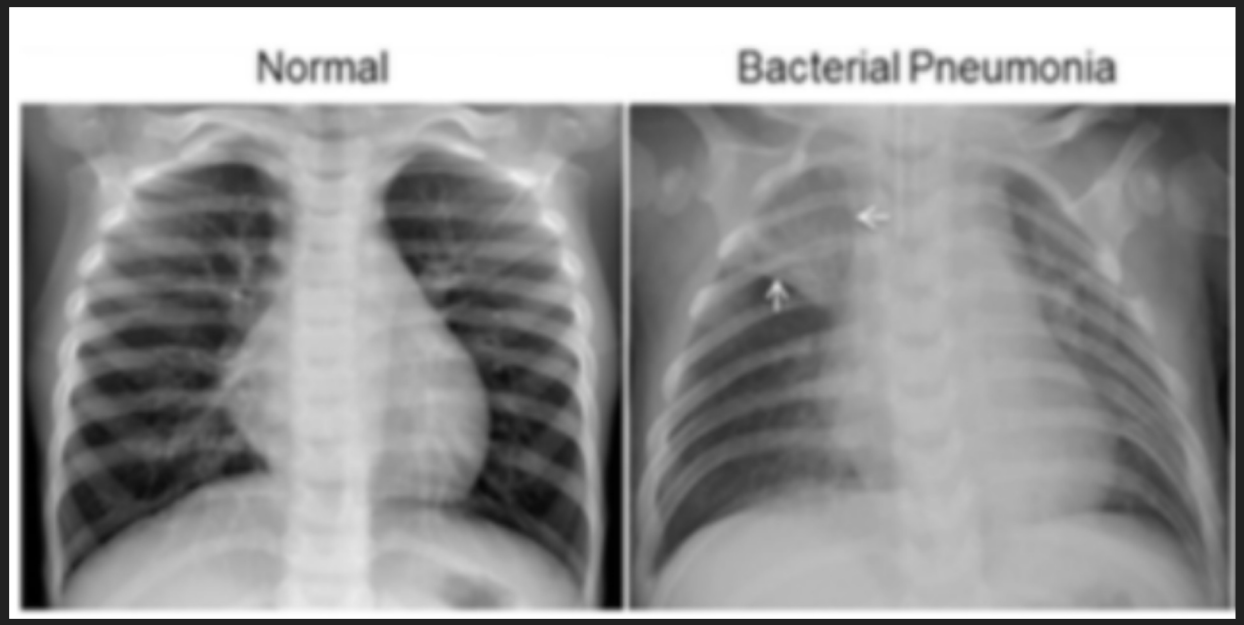
Many features less apparent

Data >>> Eyes:

Look at images first,

then look at plots

of the extracted data



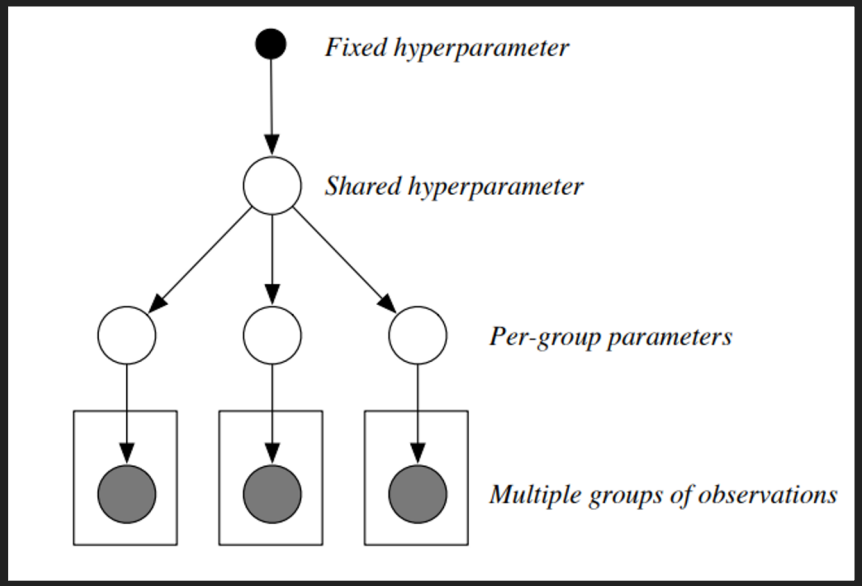
Libraries

- You don’t need to reinvent the wheel with computer vision
- If image processing is interesting to you most of the principles of feature extraction are heavily related to Computational Cameras (CS 166)
- There exist packages that can detect corners, edges, shapes, etc.
- Can be found quickly through a Google search and can also use OH to help with extracting basic information from images
- From there you can apply your knowledge and intuition to predict what extractable features would be relevant

Models / Architectures

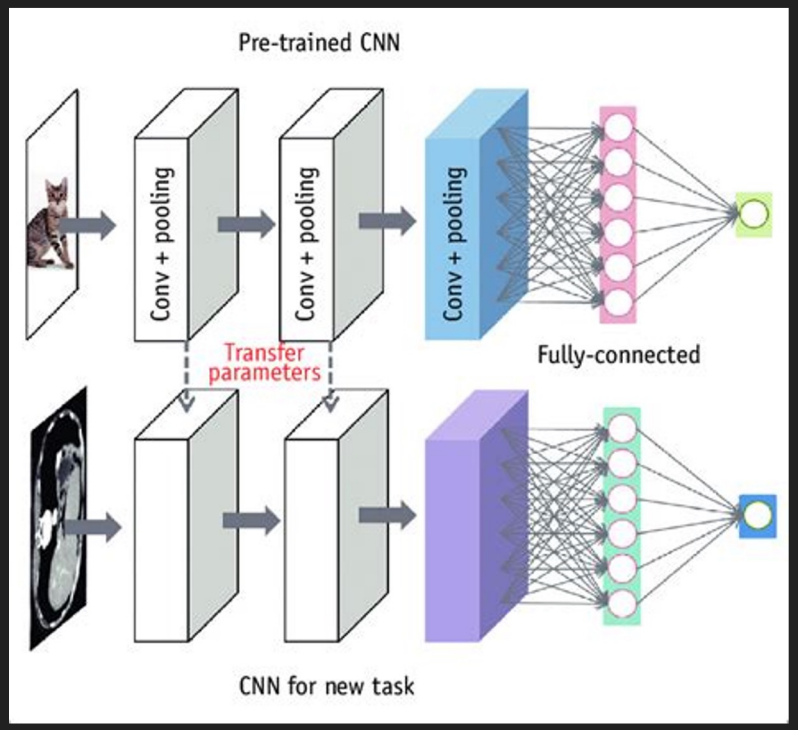
One big model…?

- Learn patterns across entire dataset, but hard to learn each pathology well?
- Ideas to do better on specific groups of data (what are the trade-offs?)
  - Train separate models for each pathology?
  - Train separate models for subpopulations of data (clustering, mixture models)?
    - For different views (frontal / lateral) ?
  - Train models for related pathologies (hierarchical models)?
- “Sharing statistical strength” (D. Blei)



Transfer learning?

- Use model pre-trained on another dataset as starting point for your task
  - Exploit patterns learned from other dataset on yours – assuming they will be relevant!
  - e.g., ImageNet? What kinds of patterns might we expect this model to have learned?
  - One view: learning a transformation from one dataset to another



Miscellaneous Tips

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- Consider storing preprocessed data to avoid redundant computations. This can save your group a lot of time, which could otherwise go into training your model(s) on more examples.
- Store the models you've used so that you can more easily form comparisons.
- Keep a record of the models you have used and in which settings they seem to work well / which settings they seem to suffer.
- Even if the models you test don't perform very well, it's great to talk about in presentations, especially if you can hypothesize why the approaches didn't work.